CANVAS 2026 Bayes Workshop Glossary of Statistical terms by Frederi Viens, 10-31-2026

- **Probability**
  - (Probability theory) The field of study for modeling the chances of this and that.
  - The chance that an event will happen, in a random event or phenomenon
  - (A probability measure): the set of all the chances of all events of interest in a random experiment or phenomenon.
    - (Axioms of probability) According to the formalism developed about 100 years ago, for a probability measure P and a set of events A, B, C, D,…
      - the probabilities of these events are numbers between 0 and 1;
      - If C is the event that A or B happen, and A and B are incompatible (they can never happe simultaneously), then P[C], the probability of C, is the sum of P[A], the probability of A, and P[B], the probability of B;
      - **In other words if A and B are disjoint, then P[A U B] = P[A] + P[B]**
      - this works for infinite sequences of events too; plus, the probability of anything at all happening (any possibility) should be 100% (i.e. = 1)
- **Random**: the state of being uncertain.
  - Random variable: a quantity whose value is uncertain and can be modeled using probability (theory).
- **Model**
  - (probability model or statistical model) A mathematical way of explaining the chances of unobserved outcomes; it has to be consistent with the axioms of probability.
  - Linear model : a mathematical model where the outcome variable Y is linearly related to the explanatory ("independent") variables X, except that one must account for statistical error by adding an "idiosyncratic'' error term ε, which is a random variable.
    - **In other words, for some constant (non-random) parameter β, we have $Y = \beta X + \varepsilon$**
  - Logistic model : (this is conceptually more complicated) a mathematical model where:
    - the outcome variable only has two possibilities (e.g. 0 or 1, Heads or Tails, Clinton or Trump, organic or non-organic, cover crop or no cover crop, loan in good standing or loan in default, etc.)
    - we model the two outcomes as having probability p or 1-p (this is because we have to be consistent with the axioms or probability)
    - the so-called log odds, i.e. $\log(p/(1-p))$, is linearly related to the independent variables, except that one must account for statistical error by adding an "idiosyncratic'' error term ε, which is a random variable.
    - **In other words,**
      - **Y is a Bernoulli random variable with success probability p,**

- **For some constant (non-random) parameter β, we have log(p/(1-p)) = β X + ε**

- **Density**: for a random variable Y which is continuous (like the future yield of a corn field, or the amount of rain that will fall on that field in July),
  - one typically tries to assume that it has a density f,
  - which means that the chance that the variable Y will fall between the values a and b is equal to the integral of the function f between a and b:
  - **in other words, when *Y* has density *f*, then P[$a < Y < b$] = $\int_a^b f(y)dy$.**
  - Examples of random variables with densities are normal (Gaussian), exponential, chi-squared, uniform, and beta random variables.
- **Discrete random variable**:
  - As opposed to a continuous random variable, that has a density, a discrete random variable is one which can only take finitely many or countably infinitely many values
  - Examples of these are Bernoulli or Binomial or Poisson or Geometric random variables.
- **Histogram**:
  - Given a bunch of numerical datapoints x_i which all measure the same quantitative thing (like the outcomes of repeating an experiment a bunch of times, or like a bunch of measurements of the same thing, over time, or over space, or over whatnot),
  - Their histogram is a picture showing how these values are distributed in bins across all the possible values of the x_i's
  - This histogram is the empirical, data-driven analogue of a density for a continuous random variable.
  - And it also makes sense for discrete data and discrete random variables.
- **Mean and standard error vs. expectation and standard deviation:**
  - the mean μ of a set of data x_i that all measure the same type of thing is their arithmetic average $\bar{x}$
  - the standard error σ of that dataset is the square root of the mean of the squared deviations $(x_i - \bar{x})^2$ between the datapoints and their mean.
  - These two concepts μ and σ are empirical statistics, based on the data.
  - In a probability model X for the data, the analogues of these statistics μ and σ are the (mathematical) expectation E[X] and the standard deviation $\sqrt{Var[X]}$ of X.
  - Unfortunately, the letters μ and σ are often used for those statistics and for their mathematical model analogues E[X] and $\sqrt{Var[X]}$.
- **Conditional probability**:
  - For an experiment or random situation with two events A and B whose probabilities are computable in principle, and so is the probability that they would occur simultaneously, the conditional probability of B given A is the probability that they would occur simultaneously divided by the probability of A.
  - This really means that we reduce the set of all possible outcomes to those that are compatible with the event A, and we recompute the probability of B assuming that A has occurred for certain.
  - In other words, $P[\ B \mid A\ ] = P[\ B\ \cap A\ ]/P[\ A\ ]$.

- **Bayesian statistics** : a statistical framework, first set up by non-confirmist English Presbyterian vicar Thomas Bayes, in which we compute conditional probabilities of the type $P[\ A \mid B\ ]$ using concepts of priors and likelihoods.
- **Prior, likelihood, and posterior** :
  - See the notation in the "Conditional probability" item.
  - In Bayesian statistics, to compute the **posterior probability** $P[\ A \mid B\ ]$,
  - one multiplies the **likelihood probability** $P[\ B \mid A\ ]$ by the **prior probability** $P[\ A\ ]$,
  - and then one divides by the probability of the observation $P[\ B\ ]$;
  - the latter can be computed by adding all the likelihood probabilities times their priors, for all possible events A_i .
  - **In other words, one has Bayes' Theorem** $\boldsymbol{P[\ A \mid B\ ] = \frac{P[\ B \mid A\ ] * P[\ A\ ]}{\sum_i P[\ B \mid A_i\ ] * P[\ A_i\ ]}}$
- **Posterior predictive probability** :
  - This is a bit more complicated to explain explicitly in full detail in a glossary,
  - But basically this is what one gets by combining the posterior probabilities of all the parameters in a model with the randomness of the idiosyncratic error, resuting in a "what if" probability model for the outcome quantity of interest (like Y) which includes uncertainty about the parameters as well as uncertainty about the original model itself.
- **Uncertainty** : this is ill-defined and can be complicated, but here are some thoughts.
  - The two types of uncertainties in the concept of posterior predictive distribution are perhaps what the "dear departed" Secretary of Defense under "W.", Donald Rumsfeld, would have liked to distinguish by calling them the **unknown unknowns (uncertainty on the parameters like β) and the know unknowns (the idiosyncratic error, like ε)**.
  - These two types of uncertainties can also be labeled as **idiosyncratic uncertainty** (as we already said) **and modeling ambiguity**.
- **Sampling**: The generic notion of sampling is like this
  - Imagine we have a random variable, like Y (or like the posterior estimate for the regression coefficient β) and we would like to generate simulated datapoints y_j whose distribution is like that of Y (or simulated values β_j whose distribution is like that of β.)
  - A method for doing this is called a sampling method.
  - The action of generating these values is called sampling.
  - These simulated datapoints (or values) are called samples.
- **Monte Carlo method, convergence**
  - The Monte-Carlo method is the basic notion of what happens when one samples from a distribution (like that of β or Y),
  - and one uses those samples to approximate mathematical statistics for those distributions (like using $\frac{1}{n}\sum_{j=1}^{n} Y_j$ or $\frac{1}{n}\sum_{j=1}^{n} \beta_j$) to represent that statistic (in these cases they would represent E[Y] or E[β] approximately).
  - The accuracy of these approximations depends on how large n is, i.e. how many samples are used, and typically goes to 0 at the same rate as $\sqrt{1/n}$ as n gets very large.

In this glossary, I also wanted to cover the concepts of :

- **Conjugacy** for Bayesian calculations,
- **posterior predictive distribution** in more detail
- how **sampling works super well for posterior predictive distributions in linear models** under simple assumptions
- an extension of the notion of sampling for Monte-Carlo methods, when the assumptions are simple but not quite simple enough to allow for directly known posterior distributions, predictive or not.
- The simplest version of the resulting extension is known as the **Gibbs sampler**, which is a simple instance of a **Markov-chain Monte-Carlo method**
- I will write this all up after we get a chance to look at some of the simple math regarding conjugacy in detail.